Simple linear regression

*with the **moderndive** package*

one explanatory variable

```
y ~ x
```

lm()
get_regression_table()
get_regression_points()
geom_smooth()

Fertility rate and life expectancy across countries

```
y ~ x
```

The **model formula**. Read the tilde as “is modeled by”. The response variable goes on the left, the explanatory variable on the right.

Response ~ Explanatory

Every modeling function in this chapter takes a formula and a data frame.

lm()

Fit a **linear regression** model: the single straight line that comes closest to every point in the cloud.

Fit a Linear Model



lm()

Example

```
life_exp_model <- lm(fert_rate ~ life_exp, data = UN_data_ch5)
```

the verb (function)

the model formula response ~ explanatory

the data frame

get_regression_table()

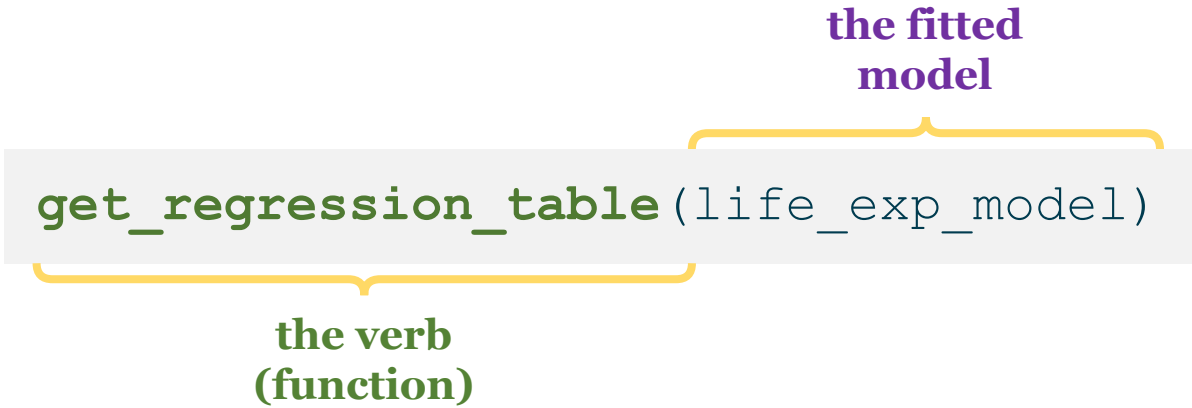
Turn a fitted model into a **tidy table**: one row per coefficient, with the intercept and the slope you interpret.

Read the Coefficients

The estimate column holds the intercept and the slope.

get_regression_table()

Example



get_regression_points()

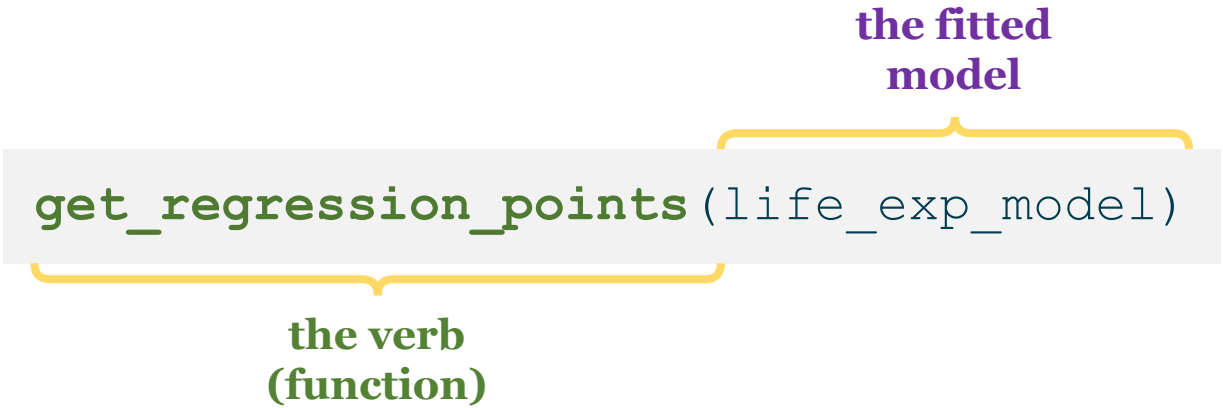
Return the **fitted value** and the **residual** for every observation, so you can see how far each point sits from the line.

Fitted Values & Residuals

residual = observed y minus fitted y-hat

get_regression_points()

Example



geom_smooth()

Draw the **regression line** on top of a scatterplot.

Example

```
ggplot(UN_data_ch5, aes(x = life_exp, y = fert_rate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

what kind of layer to add

fit a straight line

no confidence band

Interpreting

What the two numbers in the regression table actually mean.

Intercept & Slope

- The **intercept** is the fitted value when x is zero.
- The **slope** is the change in y for a one-unit increase in x.
- A **residual** is what the model missed for one observation.
- Correlation measures strength; the **slope** carries the units.